



Week 2 Cloud Security Red Teaming Report

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1. Introduction

This report documents the Week 2 tasks for the CyArt Red Teaming project. The objective was to simulate cloud security workflows, implement services using LocalStack, and demonstrate the configuration of S3, SNS, and SQS with event-driven architecture. The report includes theoretical understanding, practical execution, workflow steps, and placeholders for screenshots to ensure clarity in documentation.

2. Environment Setup

The tasks were performed in a Kali Linux environment with Python virtual environments. LocalStack was used to emulate AWS services locally. The AWS CLI was configured with dummy credentials for interaction with LocalStack.

Commands executed:

```
export AWS_ACCESS_KEY_ID=test  
export AWS_SECRET_ACCESS_KEY=test  
export AWS_DEFAULT_REGION=us-east-1
```

3. S3 Bucket Creation

Three S3 buckets were created for testing: yash-test-bucket-checkov, yash-test-bucket-logs, and yash-test-bucket-replica-checkov.

Command executed:

```
aws --endpoint-url=http://localhost:4566 s3 ls
```

```
(kali㉿kali)-[~]  
$ aws --endpoint-url=http://localhost:4566 s3 ls  
2025-09-11 02:37:25 yash-test-bucket-checkov  
2025-09-11 02:37:25 yash-test-bucket-logs  
2025-09-11 02:37:25 yash-test-bucket-replica-checkov
```



4. SNS and SQS Setup

An SNS topic and an SQS queue were created to simulate event-driven messaging. The SNS topic was subscribed to the SQS queue for event notifications.

Commands executed:

```
aws --endpoint-url=http://localhost:4566 sns create-topic --name  
s3-event-topic  
aws --endpoint-url=http://localhost:4566 sqs create-queue --queue-  
name mysqs  
aws --endpoint-url=http://localhost:4566 sns subscribe --topic-arn  
arn:aws:sns:us-east-1:000000000000:s3-event-topic --protocol sqs --  
notification-endpoint arn:aws:sqs:us-east-1:000000000000:mysqs
```

```
(kali@kali)-[~]  
$ aws --endpoint-url=http://localhost:4566 sns list-topics  
{  
  "Topics": [  
    {  
      "TopicArn": "arn:aws:sns:us-east-1:000000000000:s3-event-topic"  
    }  
  ]  
}
```

```
(kali@kali)-[~]  
$ echo "SNS event test" > test2.txt  
aws --endpoint-url=http://localhost:4566 s3 cp test2.txt s3://yash-test-bucket-checkov/ \  
--region us-east-1  
upload: ./test2.txt to s3://yash-test-bucket-checkov/test2.txt  
(kali@kali)-[~]  
$ aws --endpoint-url=http://localhost:4566 sqs receive-message \  
--queue-url http://localhost:4566/000000000000/mysqs \  
--region us-east-1  
{  
  "Messages": [  
    {  
      "MessageId": "103ccaaa-2130-40ab-b24a-68af0df61c7c",  
      "ReceiptHandle": "OTA3ZDg0NTgtY2VlZS00NmFmLWE3ZWMTMjFmYTZlZTBiZjYxIGFybjphd3M6c3FzOnVzLWVhc3QtMTowMDAwM  
DAwMDAwMDA6bXlzcXMGMTA5Y2NhYWVtMjEzMC00MGF1LWlWYnGEtNjhhZjBkZjYxYzZjIDE3NTc2MTY4MDkuMzcyMzUz",  
      "MD5OfBody": "b0cc0641a62a3756e74b39190a88e99e",  
      "Body": "{\"Type\": \"Notification\", \"MessageId\": \"aed88150-8621-4c46-b45d-baf01f017c1f\", \"TopicA  
rn\": \"arn:aws:sns:us-east-1:000000000000:s3-event-topic\", \"Message\": {\"Records\": [{\"eventVersion\":  
\": \"2.1\", \"eventSource\": \"aws:s3\", \"awsRegion\": \"us-east-1\", \"eventTime\": \"2025-09-11T18:53:17.727Z\",  
\"eventName\": \"ObjectCreated:Put\", \"userIdentity\": {\"principalId\": \"AIDAJDPLRLKLG7UEXAMPLE\"}, \"requestParameters\": {\"sourceIPAddress\": \"127.0.0.1\"}, \"re  
sponseElements\": {\"x-amz-request-id\": \"96f97e8b\", \"x-amz-id-2\": \"eftixk72aD6Ap51TnqcoF8eFidJG9Z/2\"},  
\"s3\": {\"s3SchemaVersion\": \"1.0\", \"configurationId\": \"tf-s3-topic-2025091106372639970000002\", \"bucket\": {\"name\": \"yash-test-bucket-checkov\", \"ownerIdentity\": {\"  
principalId\": \"A3NL1KOZZKExample\"}, \"arn\": \"arn:aws:s3:::yash-test-bucket-checkov\"}, \"objectKey\": \"test2.txt\", \"sequencer\": \"0055AED6DCD90281E5\", \"versionId\": \"AZk3m931xwoXjJlJC_eYLV86xic0jdZR\", \"eTag\": \"d40e654c0125d39005815f6d03b29fb3\", \"size\": 15}}}]\",  
\"Timestamp\": \"2025-09-11T18:53:17.742Z\", \"UnsubscribeURL\": \"http://localhost.localstack.cloud:4566/?Action=Un  
subscribe&SubscriptionArn=arn:aws:sns:us-east-1:000000000000:s3-event-topic:13fea93f-5c8c-4074-a15d-298ab73fe720\",  
\"Subject\": \"Amazon S3 Notification\", \"SignatureVersion\": \"1\", \"Signature\": \"QvHwyoaCs593fB0qbTtgIGbLnVG6Cf60Y2k7cJM930J6xs0R6LaKophJiQ56tZ0p77CNmR+vPLh8eFDcI2EZHj26ScAEMSfp6unnnAGL9ir63hSzi0tuA8+NHXFAwsX02BLsc2hL5VwoN4  
ee0dazEB4swM2J0f4kxyzJSeF6SiTYVFKpBfARXTTexGTxWbs8PNPKMoW4ar9JeJfipo/0tJt6Uk+hpRTu7vjH19s/M81Me/QVb9sCFrhShEdT2UwuEeif93LU/hUtKio7pBBPIA50K/woWZ4UTetI34PqT7QLPOzj+apcpWIsb9px/rW0ZCQ16G2LTiHasKc2DPAVw=\", \"SigningCertURL\": \"http://localhost.localstack.cloud:4566/_aws/sns/SimpleNotificationService-6c6f63616c737461636b69736e696365.pem\"}"  
    }  
  ]  
}
```



```
(venv)-(kali@kali)-[~/week 2]
$ aws --endpoint-url=http://localhost:4566 s3 ls
2025-09-11 02:37:25 yash-test-bucket-checkov
2025-09-11 02:37:25 yash-test-bucket-logs
2025-09-11 02:37:25 yash-test-bucket-replica-checkov

(venv)-(kali@kali)-[~/week 2]
$ aws --endpoint-url=http://localhost:4566 sns create-topic --name s3-event-topic
{
  "TopicArn": "arn:aws:sns:us-east-1:000000000000:s3-event-topic"
}

(venv)-(kali@kali)-[~/week 2]
$ aws --endpoint-url=http://localhost:4566 sqs create-queue --queue-name mysqs
{
  "QueueUrl": "http://sqs.us-east-1.localstack.cloud:4566/000000000000/mysqs"
}

(venv)-(kali@kali)-[~/week 2]
$ aws --endpoint-url=http://localhost:4566 sns subscribe ...
usage: aws [options] <command> [<subcommand> ...] [parameters]
To see help text, you can run:

aws help
aws <command> help
aws <command> <subcommand> help

aws: error: the following arguments are required: --topic-arn, --protocol

(venv)-(kali@kali)-[~/week 2]
$ aws --endpoint-url=http://localhost:4566 sns subscribe \
--topic-arn arn:aws:sns:us-east-1:000000000000:s3-event-topic \
--protocol sqs \
--notification-endpoint arn:aws:sqs:us-east-1:000000000000:mysqs
{
  "SubscriptionArn": "arn:aws:sns:us-east-1:000000000000:s3-event-topic:13fea93f-5c8c-4074-a15d-298ab73fe720"
}
```



5. Event Testing

A test file was uploaded to the S3 bucket to generate an event notification. The event was then captured using a Python script (test_s3_event.py) configured with boto3.

Command executed:

```
aws --endpoint-url=http://localhost:4566 s3 cp test2.txt s3://yash-test-bucket-checkov/
```

```
(kali@kali)-[~]
$ aws --endpoint-url=http://localhost:4566 s3api get-bucket-logging --bucket yash-test-bucket-checkov
{
  "LoggingEnabled": {
    "TargetBucket": "yash-test-bucket-logs",
    "TargetPrefix": "mybucket-logs/"
  }
}
```

```
(venv)-(kali@kali)-[~/week 2]
$ python3 test_s3_event.py
✓ Uploaded file to S3
✓ Got S3 event:
{
  "Records": [
    {
      "eventVersion": "2.1",
      "eventSource": "aws:s3",
      "awsRegion": "us-east-1",
      "eventTime": "2025-09-11T18:53:17.727Z",
      "eventName": "ObjectCreated:Put",
      "userIdentity": {
        "principalId": "AIDAJDPLRKL67UEXAMPLE"
      },
      "requestParameters": {
        "sourceIPAddress": "127.0.0.1"
      },
      "responseElements": {
        "x-amz-request-id": "96f97e8b",
        "x-amz-id-2": "eftixk72aD6Ap51TnqcoF8eFidJG9Z/2"
      },
      "s3": {
        "s3SchemaVersion": "1.0",
        "configurationId": "tf-s3-topic-20250911063726399700000002",
        "bucket": {
          "name": "yash-test-bucket-checkov",
          "ownerIdentity": {
            "principalId": "A3NL1K0ZZKExample"
          },
          "arn": "arn:aws:s3:::yash-test-bucket-checkov"
        },
        "object": {
          "key": "test2.txt",
          "sequencer": "0055AED6DCD90281E5",
          "versionId": "AZk3m931xwoXjJLjC_eyLV86xic0jdzR",
          "eTag": "d40e654c0125d39005815f6d03b29fb3",
          "size": 15
        }
      }
    }
  ]
}
```



6. Workflow Summary

1. Configure AWS CLI environment variables.
2. Create S3 buckets for testing.
3. Create an SNS topic.
4. Create an SQS queue.
5. Subscribe SQS queue to SNS topic.
6. Upload file to S3 bucket to generate event.
7. Verify event captured in SQS via Python script.

7. Conclusion

The Week 2 task successfully demonstrated cloud security event handling using LocalStack. The integration of S3, SNS, and SQS services with event-driven architecture was tested, and the events were validated using a custom Python script. This documentation provides the workflow, commands, and placeholders for screenshots required for submission.